

NFS Assignment - Questions and Answers

Question 1: Install the nfs-kernel-server package and create a shared directory named /srv/music_share.

Answer:

Install NFS Server:

```
sudo apt update  
sudo apt install nfs-kernel-server -y
```

Create Shared Directory:

```
sudo mkdir -p /srv/music_share
```

Set Permissions:

```
sudo chmod 777 /srv/music_share
```

Verify:

```
ls -ld /srv/music_share
```

Question 2: Configure /etc/exports to share /srv/music_share with read and write permissions to a specific client IP and restart NFS service.

Answer:

Edit Exports File:

```
sudo nano /etc/exports
```

Add Line:

```
/srv/music_share 192.168.1.20(rw,sync,no_subtree_check)
```

Apply Changes:

```
sudo exportfs -ra
```

Restart NFS Service:

```
sudo systemctl restart nfs-kernel-server
```

Verify:

```
sudo exportfs -v
```

Question 3: Mount the NFS share on a client machine and verify access.

Answer:

Install NFS Client:

```
sudo apt install nfs-common -y
```

Create Mount Directory:

```
mkdir -p ~/mnt/playlist
```

Mount Share:

```
sudo mount SERVER_IP:/srv/music_share ~/mnt/playlist
```

Create Test File:

```
touch ~/mnt/playlist/testfile.txt
```

Verify:

```
ls ~/mnt/playlist
```

Question 4: Add a subfolder called restaurant_menus and copy 2 PDF or image files into it.

Answer:

On NFS Server:

```
mkdir /srv/music_share/restaurant_menus
```

Copy Files:

```
cp menu1.pdf /srv/music_share/restaurant_menus/
```

```
cp menu2.pdf /srv/music_share/restaurant_menus/
```

Verify on Client:

```
ls ~/mnt/playlist/restaurant_menus
```

Expected Output:

```
menu1.pdf
```

```
menu2.pdf
```

Question 5: Unmount the NFS share, change it to read-only, remount and test access.

Answer:

Unmount Share:

```
sudo umount ~/mnt/playlist
```

Edit Exports File:

```
sudo nano /etc/exports
```

Change Entry To:

```
/srv/music_share 192.168.1.20(ro,sync,no_subtree_check)
```

Apply Changes:

```
sudo exportfs -ra
```

```
sudo systemctl restart nfs-kernel-server
```

Remount Share:

```
sudo mount SERVER_IP:/srv/music_share ~/mnt/playlist
```

Test Write Access:

```
touch ~/mnt/playlist/newfile.txt
```

Expected Result:

```
Permission denied
```

This confirms the NFS share is now read-only.